Non-Inferential State Sequence Generation (NISSG)

A Structural Theory of Ordering Dynamics and Computation-Free State Sequence Formation Based on Observational Facts

Abstract

This paper introduces **Non-Inferential State Sequence Generation (NISSG)** as a new cognitive structure that generates state sequences without relying on inference or computation, and presents **Non-Inferential State Sequence Generating Ensemble Recorder (NISSGER)** as a concrete model that satisfies this abstract structure.

The world accessible to a cognitive agent is not the physical domain itself but the **representational domain**, formed through biological sensor mappings and internal world mappings. What is mapped into this domain is not physical trajectories but instantaneous states. Each observed state carries the factual guarantee that "a non-breaking transition existed in the physical domain," and the juxtaposition of such facts forms an **ordering of non-breaking transitions** that regulates the connectivity between states. This ordering is not derived from equations or inference rules but is an abstract structure externally supported by the reproducibility of the physical domain.

NISSG demonstrates that state sequences arise from a remarkably simple dynamics:
juxtaposition of observational facts → ordering → next-state selection → sequence formation.
This structure eliminates the need for transition-validity computation, inference-rule application, or symbolic-consistency checking, yielding a structural advantage described as **"computation cost evasion."**

NISSGER implements this abstract structure in minimal form and functions as an **Ensemble Recorder** that realizes the minimal core of intelligence—**recording, reactivation, chaining, and composition**—without computation. It exhibits a decisive advantage: partial observations can reactivate entire recorded sequences, enabling phenomena such as grounding, imitation, association, and sequence composition without inference.

However, NISSGER is inherently passive and does not perform active generation, exploration, or internal reorganization. Its ability to raise action sequences from partial observations implies potential safety concerns in certain applications. Although the author does not intend to publish active sequence-generation models, others may eventually implement such systems. For this reason, the development of an **ethical and structural framework (ethics model)** for ensuring the safety of higher-order models becomes an important future research direction.

Furthermore, this paper identifies a structural vulnerability: the quality of state-sequence generation depends entirely on the quality of sensor mappings. This **sensor dependence** suggests a new research trajectory in which the quality of intelligence is determined not by computational sophistication but by the sophistication of observation.

NISSG and NISSGER are not models for explaining the complexity of intelligence; rather, they extract the **minimal core** that makes such complexity possible. Their surprisingly simple structure is itself evidence of the correctness of computation-free cognitive modeling.

1. Problem Statement:

The Non-Computational Establishment of Validity Arising from Reproducibility Structures

Cognitive phenomena such as known/unknown discrimination, inference, symbol manipulation, and linguistic operations have traditionally been explained on the assumption that some form of computation or inferential process is required.

However, as argued in *Reproducibility as a Structural Basis for Cognition*
(Zenodo: 10.5281/zenodo.23259601),
the world an agent actually handles is not the physical domain itself but the **representational domain**, formed through biological sensor mappings and internal world mappings.

What is mapped into the representational domain is not physical trajectories but only instantaneous **states** of structural entities.
The agent treats these states as entities within the world, yet the causal structure of their transitions is not mapped.
Consequently, reproducibility within the representational domain is limited to a juxtaposition of fragments of reproducibility from the physical domain—
a condition we call **partial reproducibility**.

This structural condition has a decisive consequence.
An observed state already carries validity by virtue of the fact that it "occurred as a non-breaking transition in the physical domain at some point in the past."
Thus, the agent does not need to reconstruct causal structure internally or derive correctness through higher-order symbolic computation.

The crucial point is that:

the agent does not lack computational ability;
rather, the cognitive structure itself is fundamentally non-computational.

Validity is externally guaranteed by the reproducibility structure of the physical domain,
so the agent need not expend computational resources to verify it.
This structural advantage is referred to in this paper as **computation-cost evasion**.

Purpose of This Paper: Introducing NISSG (Theory) and NISSGER (Model)

The purpose of this paper is to extend the structural advantage of computation-cost evasion and to:

(1) Formalize Non-Inferential State Sequence Generation (NISSG)
as an abstract theory of state-sequence formation,

(2) Construct Non-Inferential State Sequence Generating Ensemble Recorder (NISSGER)
as a concrete model that satisfies this abstract theory.

NISSG is grounded in:

- the constraints of the representational domain,
- the structural fact that only states are mapped,

- the validity conferred by observational facts.

Based on these premises, NISSG provides
an abstract structure for state-sequence generation that requires neither inference nor computation.

NISSGER is designed as a concrete architecture that satisfies this abstract structure.
Its essence is:

a composite recorder capable of handling state sequences
without equations, inference rules, or model-specific formulations.

An agent forms “known” states through the following dynamics alone:

- sensor mapping,
- ordering of non-breaking transitions,
- updates driven by ordering dynamics,
- reactivation (chaining),
- formation of state sequences.

This structure eliminates the need for:

- designing computational equations,
- applying inference rules,
- verifying validity,
- revising model formulations.

In other words, NISSGER is:

a recorder that can store and reproduce complex state sequences
without computation.

This paper first presents NISSG as an abstract theory,
and then constructs NISSGER as a concrete model that satisfies it.

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2. Theoretical Framework:

The Abstract Structure of State Sequence Generation Based on Observational Ordering Dynamics

2.0 Introduction: Positioning NISSG

State sequence generation is an abstract structure that enables an agent—one that cannot access causal structure—to handle the world solely through the reactivation of observed states.

As shown in the previous paper (Zenodo: 10.5281/zenodo.23259601),
the only world accessible to the agent is the **representational domain**,
formed through biological sensor mappings and internal world mappings.
What is mapped into this domain is not physical trajectories but instantaneous **states**.
The agent treats these states as entities within the world,
yet neither trajectories nor the causal structure behind state transitions are mapped.

Under this constraint, each observed state carries the factual guarantee that
“a non-breaking transition existed in the physical domain.”
The juxtaposition of such facts forms an **ordering of non-breaking transitions**
that regulates the connectivity between states.

This ordering is not derived from equations or inference rules;
it is an abstract structure formed through the juxtaposition of observational facts,
unconstrained by any specific procedure.

This chapter introduces this abstract ordering structure as
Non-Inferential State Sequence Generation (NISSG)
and clarifies the general framework under which state sequence generation is established.

2.1 General Framework of State Sequence Generation : Juxtaposition of Non-Breaking Transitions

Each observed state carries the factual property that
it resulted from a non-breaking transition in the physical domain.
The juxtaposition of these facts forms an ordering that regulates
how easily states connect to one another.

State sequence generation arises as an abstract dynamics grounded in this ordering.

1. **The sensor observes a state**
   States in the representational domain are not physical trajectories
   but discrete entities obtained through sensor mappings.

2. **Each observed state carries the fact of a non-breaking transition**
   This fact serves as the structural basis for regulating relationships among states.

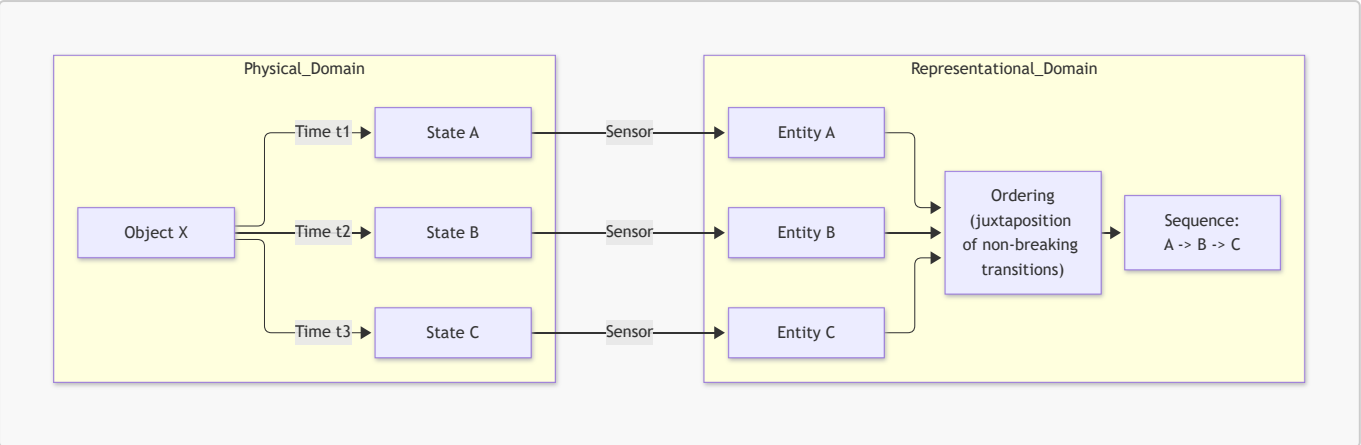
3. **Juxtaposition of non-breaking transitions forms an ordering**
   Observed states are naturally juxtaposed based on the existence of non-breaking transitions,
   forming an abstract structure that determines their “connectability.”

4. **Next states are selected according to the ordering**
   Highly connectable states are more easily reactivated,
   and structural variability expands the possibility space.
   Next-state selection arises naturally within the ordering, not through inference.

5. **Chains of reactivation become state sequences**
   A state sequence is not a reconstruction of causal structure
   but a sequence of reactivations within the representational domain.

The possibility space defined by this ordering functions as the agent’s
drive domain.
Its reversibility or irreversibility depends on sensor characteristics
and the structure of the ordering.

Fig1. Schematic flow of NISSG from physical transitions to representational sequencing.



2.2 Ordering as an Abstract Structure Independent of Specific Formalisms

Ordering in state sequence generation is not a fixed rule or formal procedure.
It is an abstract structure arising from the juxtaposition of observed states.

Ordering regulates the connectability among states
without being constrained by any specific computational or symbolic procedure.
Its concrete form varies depending on the nature of the system.

Examples of ordering dynamics include:

- temporal proximity,
- co-occurrence frequency,
- structural similarity within the representational domain,
- constraints imposed by symbolic systems or context.

These are merely examples;
the general framework presented in this paper accommodates all of them.

The purpose of this section is not to restrict ordering to a particular form,
but to demonstrate that state sequence generation arises under
abstract ordering with minimal formal constraints.
Which form is adopted depends on the goals and constraints of the system,
and NISSG does not impose restrictions on that choice.

2.3 Independence from Computational Cost: The Structural Advantage of Computation-Free Ordering

State sequence generation in NISSG does not rely on equations or inference rules.
Transition validity is not computed;
it is supported by the ordering naturally formed through the juxtaposition of observed states.

This structure differs fundamentally from existing cognitive or machine-learning models.
Traditional models explicitly define transition validity through equations
and require human designers to construct, adjust, and evaluate these formulations.
Thus, model validity always depends on the consistency of the equations,
and the cognitive cost of designing and verifying them cannot be avoided.

NISSG eliminates this requirement.
State sequences arise naturally from:

- observed non-breaking transitions,
- the ordering formed from them,
- the possibility of reactivation.

The agent does not “compute transitions”;
it simply **rides the ordering of non-breaking transitions**.

This structure evades the following computational costs:

- computing transition validity,
- designing equations,
- verifying equation consistency,
- revising equations,
- applying inference rules,
- checking symbolic consistency.

All of these are replaced by the juxtaposition of observational facts.

2.4 Essence of State Sequence Generation:Ordering of Non-Breaking Transitions × Next-State Selection

The essence of state sequence generation lies in the combination of:

- the ordering formed through the juxtaposition of observed states, and
- next-state reactivation within the possibility space defined by that ordering.

Each observed state carries the factual guarantee of a non-breaking transition,
and this fact regulates the connectability among states.
Ordering is not derived from equations or inference rules;
it is an abstract structure arising from observational juxtaposition.

The agent selects the next state based on this ordering.
Next-state selection is not inferential or computational;
it arises through chains of reactivation within the ordering’s possibility space.

This structure—**ordering × next-state selection**—constitutes the abstract dynamics
of state sequence generation and provides the foundation for implementing
the external validity guaranteed by reproducibility structures
as a cognitive model.

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3. Minimal Implementation: Verifying the Operation of the Abstract Structure

This chapter presents the simplest possible minimal model to verify how the abstract structure introduced in Chapter 2 operates in practice.

This model is not the full version of NISSGER.
Rather, it is the minimal implementation that satisfies the abstract structure of NISSG:

juxtaposition of observational facts → ordering → next-state selection

The complete code for this minimal model is available in the GitHub repository:

https://github.com/SwayMagpie/NISSG_Theory

This chapter focuses only on the structure and behavior of the model.

3.1 Model Structure

The minimal model consists of the following three components:

- **Observation function** `observe(state)`
  Records each observed state and accumulates the transition frequencies
  for "previous state → current state."
  This corresponds to the "juxtaposition of observational facts" described in Chapter 2.

- **Transition matrix** `transition_matrix`
  Formed solely from observed transition frequencies,
  without any computational equations.
  This corresponds to "ordering unconstrained by specific formalisms."

- **Next-state selection** `simulate_chain(start_state)`
  Reactivates the most frequent transition from the current state.
  Next-state selection arises naturally within the possibility space defined by ordering,
  not through inference or computation.
  This corresponds to the structure of "ordering × next-state selection."

3.2 Condition: Generating a Context-Dependent Environment

The model cannot observe context (hidden states).
Therefore, we generate an environment in which transition tendencies change depending on context,
and examine how the model records such an environment as "mere transition sequences."

In this minimal model, we prepare an environment where
two contexts (A/B) switch randomly,
and the transition rules differ between contexts.

The design intention is as follows:

- Because the causal structure of context cannot be observed,
  the model can only treat a "context-dependent complex environment"
  as a simple sequence of observed states.

- Nevertheless, juxtaposition of observational facts still forms ordering,
  enabling next-state selection.

- This demonstrates that the abstract structure of NISSG
  does not break even in environments with hidden states.

3.3 Example Output (Excerpt)

Below is an excerpt from the output produced by running `minimal.py`.

- **Estimated transition matrix (excerpt)**

```
warm → {'cool': 9, 'hot': 5}
cool → {'cold': 9, 'warm': 5}
```

- **Generated sequence (beginning only)**

```
['warm', 'cool', 'cold', 'hot', 'warm', 'cool', ...]
```

- **Match ratio**

```
14/49 matches
```

- **Next-state reactivation**

```
predict_next('warm') → 'cool'
```

These results demonstrate that even with a minimal structure,
the abstract dynamics introduced in Chapter 2 operate exactly as intended.

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4. Conditions and Constraints:Sensor Dependence as the Structural Vulnerability Determining the Quality of the Representational Domain and Future Research Directions

This chapter examines the conditions under which the **abstract structure of state sequence generation (NISSG)** introduced in Chapter 2 is established, and the constraints under which the minimal implementation (NISSGER) presented in Chapter 3 operates.

NISSG is founded on an extremely simple structure:

**states obtained through sensor mappings are juxtaposed,
and their connectability is updated as ordering.**

The agent cannot access trajectories or causal structure within the representational domain;
it can only handle observed states.

Thus, the fact that a state was observed guarantees that
"the transition leading to that state did not break in the physical domain,"
functioning as a **structural validity certificate**.

This structure enables the formation of known states solely through the juxtaposition of observational facts,
without requiring complex computation or inference—
a major structural advantage.

However, this advantage contains a critical vulnerability:
it depends entirely on the quality of sensor mappings.

This chapter clarifies how sensor dependence determines the quality of the representational domain,
the quality of known states, and the precision of sequence generation,
and how this constraint shapes future research directions.

Note on *kichi* (known states)

The term *kichi* used in this paper follows the definition introduced in
Reproducibility as a Structural Basis for Cognition
(Zenodo: 10.5281/zenodo.23259601),
referring to states that can be internally reactivated by the agent.

4.1 Sensor Quality Determines the Quality of the Representational Domain

The world an agent handles is not the external world itself,
but the **representational domain** formed through biological sensor mappings
and internal world mappings.

States in the representational domain are the result of these mappings,
and their properties are determined by how sensors extract information—
specifically:

- spatial resolution (granularity of spatial distinctions),
- temporal resolution (sensitivity to temporal changes),
- multidimensionality (variety of physical quantities captured),
- integrative structure (degree of multimodal coupling),
- reactivation stability (consistency of state reproduction).

Higher sensor quality yields richer representational states
and higher-quality known states formed through state sequence generation.
Conversely, coarse sensors produce coarse representational domains
and lower-quality known states.

Thus:

> **The quality of known states is determined by the quality of sensors.**

This is the most important condition for the abstract structure of state sequence generation,
and simultaneously its structural vulnerability.

4.2 The Precision of State Sequence Generation Also Depends on Sensor Quality

Section 4.1 established that the static quality of representational states
is determined by sensor characteristics.

This section examines how sensor quality affects the **dynamic** process
of state sequence generation.

State sequence generation links observed states through ordering
and forms internally reactivatable sequences.
The precision and reproducibility of this process depend on
how finely and how stably sensors provide states.

Specifically, sensor characteristics directly determine:

- **Sequence resolution**
  Finer state granularity yields higher-resolution sequences.
- **Prediction and regeneration accuracy**
  Stability of past transition patterns depends on the reproducibility of sensor input.
- **Adaptation to environmental change**
  Higher sensitivity and frequency of state updates enable faster adaptation.

These structural consequences show that:

> **The effectiveness of sequences generated by this model**
> **depends entirely on sensor quality.**

The quality of states is determined by sensors,
and the quality of sequences formed from those states
is likewise determined by sensors.

This is the dynamic establishment condition of the abstract structure.

4.3 Modern Artificial Sensors Cannot Reproduce the Structure of Biological Sensors (Current Limitation)

The dependence of state sequence generation on sensor quality
was established in Sections 4.1 and 4.2.

However, a decisive reality must be acknowledged:
modern artificial sensors cannot reproduce the structural properties of biological sensors.

Biological sensors—vision, audition, touch, vestibular sense, proprioception, interoception—
are multidimensional, high-resolution, and highly integrated.
They are not mere “input devices,”
but complex structures with density, diversity, coupling, and reactivation stability
that artificial sensors cannot match.

For example, bird feathers contain dense arrays of mechanosensory filaments
that detect local pressure, fluid dynamics, and posture simultaneously,
enabling precise motor control within a richly integrated representational domain.
Artificial sensors achieve only a tiny fraction of this density and coupling.

Thus:

> **Modern science cannot reproduce the structural properties of biological sensors.**

This paper abstracts sensor quality to maintain theoretical implementability,
but such abstraction is a **structural evasion**,
not a reproduction of biological sensor architecture.

This limitation is the most critical structural vulnerability
in the establishment conditions of state sequence generation.

4.4 Sensor Dependence Is Not a Weakness but a Direction for Future Research

Although modern sensors cannot reproduce biological sensor structure,
this limitation is not a weakness of the model.
Rather, it is a **structural consequence that indicates the future direction of research**.

Historically, the advancement of knowledge has been inseparable from
the advancement of observational instruments:
telescopes for astronomy,
microscopes for biology.

Likewise, cognitive science, AI, and robotics
should evolve alongside sensor development.

State sequence generation forms known states
through reactivation and ordering of observational facts,
not through computation or inference.
Therefore:

> **Sensor quality determines the quality of intelligence.**

This perspective provides a new foundation for intelligence research,
offering a unified explanation for phenomena such as
the boundary between known and unknown,
sequence formation,
environmental adaptation,
and prediction.

Sensor dependence is not a limitation of the model;
it is a structural indicator of where intelligence research must go next.

4.5 Conclusion: Sensor Research Is the Future Direction of Intelligence Research

The state sequence generation model presented in this paper
is founded on the extremely simple structure of sensor mapping
and ordering dynamics.
This structure enables the formation of known states
through reactivation and ordering alone,
without complex computation or inference.

However, this advantage depends entirely on sensor quality.
The quality of the representational domain,
the precision of sequence generation,
adaptation to environmental change,
and stability of prediction
are all determined by how richly sensors provide states.

As shown in previous sections,
modern science cannot reproduce biological sensor structure.
This may appear to be a weakness,
but it is in fact a structural consequence indicating
the future direction of intelligence research.

Just as telescopes transformed astronomy
and microscopes transformed biology,
cognitive science, AI, and robotics
will advance through the development of sensors.

State sequence generation suggests that
the sophistication of observation—not computation—determines the quality of intelligence.

Thus:

> **Sensor quality determines the quality of intelligence.**

This structural fact defines the future trajectory of intelligence research.
The model presented here provides a theoretical foundation for that direction.

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5. What NISSG and NISSGER Provide:A Structural Interpretation of Essence, Advantages, Limitations, and Necessity

Up to Chapter 4, we established that the abstract structure of state sequence generation (NISSG)
is founded on the extremely simple dynamics of sensor mapping and ordering,
and that its establishment conditions depend entirely on sensor quality.

The minimal implementation presented in Chapter 3 (NISSGER)
realizes this abstract structure in its simplest form,
recording, reactivating, and chaining state sequences
without inference or computation.

The simplicity of “handling state sequences without inference”
is not a weakness but the result of extracting the **minimal core of intelligence**.

This chapter reinterprets the structures provided by NISSG and NISSGER
at a higher level of abstraction,
showing that their essence can be understood as an
Ensemble Recorder.

We then organize the **essence, advantages, limitations, and necessity**
of this Ensemble Recorder,
clarifying its position within the hierarchical structure of intelligence models.

5.1 Essence: NISSGER Is an “Ensemble Recorder”

The essence of NISSGER lies in its ability to handle state sequences
without inference or computation.
The agent records states obtained through sensor mappings,
reactivates them, and chains them to form known states.
When all computational equations and inference rules assumed by traditional models
are removed, the agent performs only the following simple dynamics:

- acquisition of states through sensor mapping,
- ordering of non-breaking transitions,
- updates driven by ordering dynamics,
- reactivation (chaining),
- formation of state sequences.

Abstracting this structure reveals that NISSGER is not merely a recorder.
It is a higher-order recorder that includes not only
“recording and replaying states”
but also
“chaining reactivations and composing sequences.”

The autonomous interaction of sensor mapping, ordering, reactivation,
and compositionality—without reduction to a single computational equation—
is precisely what defines an **Ensemble Recorder**.

Thus, NISSGER is:

> **an Ensemble Recorder that records, replays, chains, and composes complex state sequences without computation.**

The “ER” in NISSGER derives from this structural abstraction,
accurately reflecting the model’s essential nature.

5.2 Advantage: A “Compositional Recorder” That Chains from Partial Observation

The greatest advantage of NISSGER is:

> **it can chain from partial observation.**

This structural benefit does not exist in traditional computation-based sequence models.
NISSGER can take a fragment of an observed state
and reactivate the entire recorded sequence.

This yields several advantages:

- ✓ **Partial observation → full sequence reactivation**
  Association arises without computation.
- ✓ **Partial observation → chaining of representational states**
  Natural grounding emerges.
- ✓ **Partial observation → chaining of action states**
  Natural imitation emerges.
- ✓ **Partial observation → compositional transformation of sequences**
  Recorded sequences combine to produce new structures.

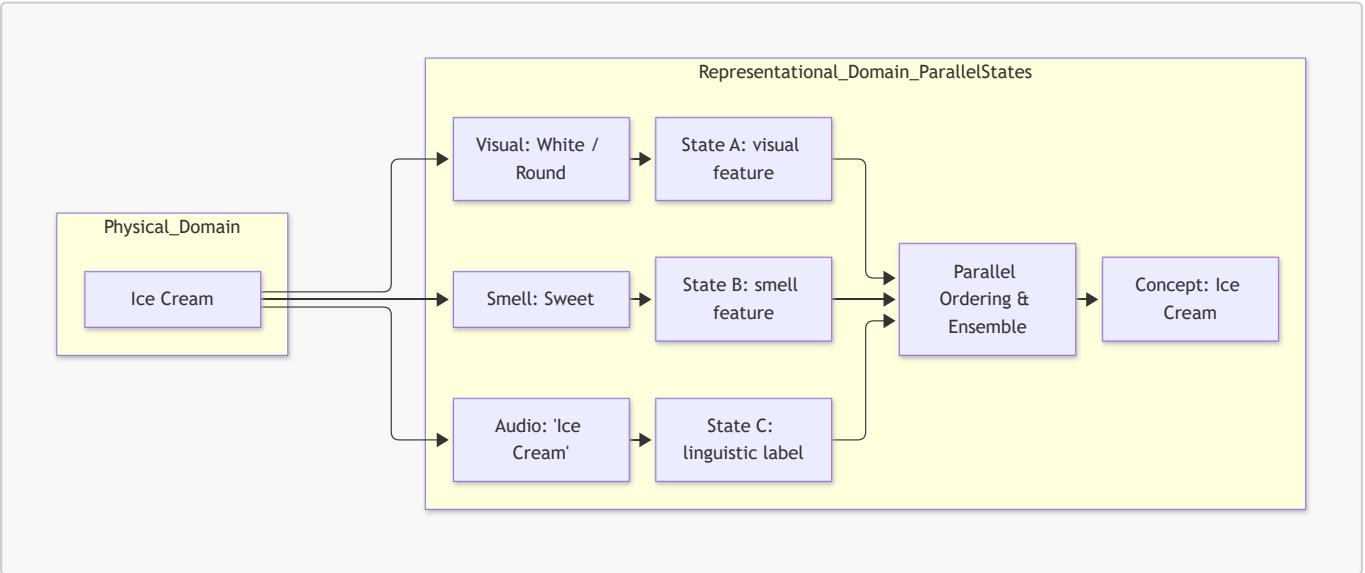
Thus, NISSGER is:

> **a compositional recorder in which recorded sequences**
> **rise and chain from partial observation.**

This structural advantage enables “association”
without the computational cost required by traditional models.

Furthermore, this structure naturally extends
to higher-order variants (NISSGER-Hebb, NISSGER-Chem),
which support more complex chaining, composition, and reactivation.
Although this paper does not address their implementation,
NISSGER provides the foundational structure for such advanced behavior.

Fig2. Partial observation of multimodal states (vision, smell, sound) reactivates the entire recorded sequence, yielding natural grounding.



5.3 Limitation: It Cannot Generate (Passive, Not Active)

NISSGER is the minimal core of intelligence,
but not the whole of intelligence.

It does **not** perform:

- generation of new state sequences,
- exploration of the unknown,
- reorganization of internal world models,
- reconstruction of symbolic structures,
- inference,
- algorithm generation.

Thus, NISSGER is:

> **passive, not generative.**

This is not a weakness but a structural position
within the hierarchy of intelligence models.
NISSGER provides the minimal core—recording, replay, chaining, composition—
while active generation and reorganization belong to higher-order models
built on top of this core.

However, despite being passive,
NISSGER possesses the property that:

> **action sequences can rise and chain from partial observation.**

This structure forms the basis for imitation and action chaining,
and therefore requires safety considerations in certain applications.

This paper does not address generative models.
The author does not intend to publish active sequence-generation models,
but acknowledges that others may eventually implement them.
Thus, the development of an
ethical and structural framework (ethics model)
to ensure the safety of such higher-order models
is an important future research direction.

5.4 Necessity: A Recorder Is a Necessary Condition for Intelligence

NISSGER is not the whole of intelligence,
but it is a **necessary condition** for intelligence.

The reasons are clear:

- ✓ What cannot be recorded cannot be replayed.
- ✓ What cannot be replayed cannot form sequences.
- ✓ What cannot form sequences cannot distinguish known from unknown.

- ✓ What cannot distinguish known from unknown cannot be intelligent.

Thus:

> **A recorder is a necessary condition for intelligence.**

Importantly, the minimality of this recorder
is not a reason to dismiss it.
Rather, **its minimality is evidence of necessity**.

Regardless of how complex an intelligent system becomes,
it must contain:

- state capture,
- reactivation,
- chaining,
- composition.

Any structure lacking this core
cannot establish the known/unknown distinction
and therefore lacks the foundation of intelligence.

NISSGER presents this minimal core
in a **composite, low-cost, and compositional** form,
distinguishing it decisively from traditional models.

5.5 The True Value of NISSGER:Extracting the Minimal Core of Intelligence (Record → Reactivate → Chain → Compose)

The true value of NISSGER lies in reducing the minimal core of intelligence
to the following four steps:

1. **Record (state capture)**
2. **Reactivate (reactivation)**
3. **Chain (sequence continuation)**
4. **Compose (sequence merging)**

These four steps are structural necessities
present in every intelligent system,
regardless of complexity.

NISSGER's contribution is to extract these four steps
as a **pure minimal structure that operates without computation**.

Thus, NISSGER does not explain the complexity of intelligence;
it extracts the **minimal core that makes complexity possible**.

This extraction provides a **structural advantage**
that computational models could not reach.

By demonstrating that
record → reactivate → chain → compose
can be realized without inference or computation,
NISSGER successfully presents the foundational dynamics of intelligence
in their minimal form.

5.6 Conclusion: The “Surprising Simplicity” of NISSGER Is Correct

NISSGER is not the whole of intelligence,
but it is a necessary condition for intelligence.
Its surprising simplicity is not a weakness
but a sign of structural correctness.
Simplicity is the foundation upon which more complex structures arise.

This paper presented the Ensemble Recorder
as the minimal core of intelligence—
recording, reactivation, chaining, composition.
Any structure lacking this core
cannot support intelligence,
regardless of how complex its generative or inferential mechanisms may be.

Higher-order variants of NISSGER
(generation, reorganization, active sequence formation)
will not be published by the author.
These models involve action-sequence formation
and the rise of action chains from partial observation,
requiring careful ethical and safety considerations.
Although this paper does not address their details,
the development of an
ethical and structural framework (ethics model)
for such higher-order models
is an important future research direction.

NISSGER is not a model for explaining the complexity of intelligence;
it is a structure that extracts the minimal core
that makes such complexity possible.
Its simplicity is the true value of the theory.
